SkillGauge AI – Assessment Report

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MCQ Section (1 Mark Each)

1. Which of the following is the primary language used in the NPTEL - Python for Data Science course?
A) R
B) Java
C) Python
D) SQL

Student Answer: C) Python
Correct Answer: C) Python
Marks: 1 / 1

2. What does AR stand for in the context of Reality Defining technologies mentioned?
A) Augmented Reality
B) Artificial Reality
C) Advanced Rendering
D) Automated Reality

Student Answer: C) Advanced Rendering
Correct Answer: A) Augmented Reality
Marks: 0 / 1

3. Which tool is mentioned for developing interactive dashboards in the profile?
A) VS Code
B) Github
C) Dataiku and Spotfire
D) BigQuery

Student Answer: C) Dataiku and Spotfire
Correct Answer: C) Dataiku and Spotfire
Marks: 1 / 1

4. What is the primary focus of the 'Trust and Security with Google Cloud' certification?
A) Web Development
B) Machine Learning
C) Cloud Security
D) Data Analysis

Student Answer: B) Machine Learning
Correct Answer: C) Cloud Security
Marks: 0 / 1

5. Which programming language is used for the front-end development mentioned in the profile?
A) Python
B) CSS
C) SQL
D) Java

Student Answer: C) SQL
Correct Answer: B) CSS
Marks: 0 / 1

6. In the context of the internship at Polenza Tech Solutions, what database technology was explored?

A) PostgreSQL
B) MongoDB
C) MySQL
D) SQLite

Student Answer: B) MongoDB
Correct Answer: B) MongoDB
Marks: 1 / 1

7. Which machine learning algorithm type is implied in the 'Bank Customer Churn Prediction' project?
A) Clustering
B) Regression
C) Classification
D) Dimensionality Reduction

Student Answer: B) Regression
Correct Answer: C) Classification
Marks: 0 / 1

8. The 'Diabetic Retinopathy Prediction Using Xception' project likely involves which domain of AI?
A) Natural Language Processing
B) Computer Vision
C) Reinforcement Learning
D) Time Series Analysis

Student Answer: C) Reinforcement Learning
Correct Answer: B) Computer Vision
Marks: 0 / 1

9. Which organization offered the GET Internship where predictive models were designed?
A) Polenza Tech Solutions Pvt Ltd
B) SRM Valliammai Engineering College
C) Renault Nissan Technology & Business Centre India
D) ISTE

Student Answer: D) ISTE
Correct Answer: C) Renault Nissan Technology & Business Centre India
Marks: 0 / 1

10. What is the core skill emphasized in the career objective regarding data?
A) Data Visualization
B) Data Cleaning and Pattern Detection
C) Database Administration
D) Network Security

Student Answer: D) Network Security
Correct Answer: B) Data Cleaning and Pattern Detection
Marks: 0 / 1

11. The project 'OTOGALUCOSENSE : AI-Powered System For Glaucoma And Otitis Media Detection' most directly relates to which application area of AI?
A) Finance
B) Healthcare
C) Education
D) E-commerce

Student Answer: B) Healthcare
Correct Answer: B) Healthcare
Marks: 1 / 1

12. Which of the following is a listed personal skill that is NOT a programming language?

A) Python
B) SQL
C) Leadership
D) HTML

Student Answer: B) SQL
Correct Answer: C) Leadership
Marks: 0 / 1

13. What does MR stand for in the context of Reality Defining technologies?
A) Mixed Reality
B) Mobile Reality
C) Master Reality
D) Modified Reality

Student Answer: B) Mobile Reality
Correct Answer: A) Mixed Reality
Marks: 0 / 1

14. In the context of ML Pipeline development, what is a key phase mentioned in the profile?
A) Data Entry
B) Model Training & Evaluation
C) User Interface Design
D) Hardware Procurement

Student Answer: C) User Interface Design
Correct Answer: B) Model Training & Evaluation
Marks: 0 / 1

15. Which of the following is NOT a listed certification or course in the profile?
A) NPTEL - Python for Data Science
B) Great Learning - Front End Development - CSS
C) Coursera - Machine Learning Specialization
D) Simplilearn - Trust and Security with Google Cloud

Student Answer: D) Simplilearn - Trust and Security with Google Cloud
Correct Answer: C) Coursera - Machine Learning Specialization
Marks: 0 / 1

Descriptive Section (2 Marks Each)

Q: Explain the fundamental differences between Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR). Provide one distinct application for each.
Answer: Augmented Reality (AR): Enhances the real world by overlaying digital content onto it. Application: Navigation apps showing directions on live camera view. Virtual Reality (VR): Completely immerses users in a fully virtual environment. Application: VR-based medical or flight training simulations. Mixed Reality (MR): Combines real and virtual elements that interact in real time. Application: Industrial maintenance using interactive holograms for repair guidance.
Evaluation: Here's an evaluation of the provided answer: **Score: 2/2** **Reason:** The answer correctly identifies the fundamental differences between AR, VR, and MR by describing the core concept of each. It also provides a distinct and appropriate application for each technology, clearly illustrating their use cases. The explanations are concise and accurate.

Q: Based on your understanding of the NPTEL - Python for Data Science course and your career objective, describe a typical data science workflow for a new project. Mention at least three key steps.
Answer: A typical data science workflow starts with understanding the problem and collecting relevant data based on the project objective. Next comes data cleaning and exploratory data analysis (EDA) to understand patterns, followed by model building, evaluation, and result interpretation to support decision-making.

Evaluation: Here's an evaluation of the provided answer: **Score: 2/2** **Reason:** The answer accurately outlines a typical data science workflow and mentions at least three key steps: 1. **Understanding the problem and collecting relevant data:** This encompasses the initial problem definition and data acquisition phases, which are crucial for any project. 2. **Data cleaning and exploratory data analysis (EDA):** This highlights the essential pre-processing and understanding steps to make the data usable and gain initial insights. 3. **Model building, evaluation, and result interpretation:** This covers the core analytical and predictive phases, leading to actionable insights. The answer is concise, directly addresses the question, and aligns with the concepts likely covered in a course like "Python for Data Science" where these fundamental workflow stages are taught. It demonstrates a good understanding of the process.

Q: Elaborate on the importance of 'Model Training & Evaluation' within an ML Pipeline. What are some common metrics used for evaluating a classification model?
Answer:
Evaluation: Score: 2/2 Reason: The answer comprehensively addresses both parts of the question. It clearly elaborates on the importance of "Model Training & Evaluation" within an ML pipeline, explaining its role in model development and improvement. It also provides a good list of common metrics used for evaluating classification models, with brief explanations of what they measure. The structure is logical and easy to follow.

Q: Discuss the potential applications of the AI-powered systems for Glaucoma and Otitis Media detection in a real-world healthcare setting. What challenges might be faced during deployment?
Answer: AI-powered systems for Glaucoma detection can assist doctors by analyzing eye images to detect early signs of vision loss, enabling timely treatment and reducing blindness risk. Similarly, Otitis Media detection systems can help identify ear infections from medical images or signals, supporting faster diagnosis in clinics and rural healthcare centers. During deployment, challenges may include data privacy and security, limited availability of high-quality labeled medical data, and gaining clinical trust and regulatory approval. Additionally, model performance may vary across populations and devices, requiring continuous monitoring and updates.
Evaluation: Score: 1.5 Reason: The answer provides a good overview of the potential applications for both Glaucoma and Otitis Media detection. It also touches upon several relevant challenges during deployment. However, it could be more detailed in discussing *how* these applications would function in a real-world setting (e.g., integrated into existing workflows, impact on patient throughput) and elaborating on the nuances of each challenge (e.g., specific data privacy concerns for medical data, the "black box" problem for clinical trust). The "real-world healthcare setting" aspect could be explored more deeply.

Q: Considering your experience with Flask and MongoDB, how would you approach building a simple RESTful API for a web application? Briefly outline the key components involved.
Answer: I would use Flask to define REST endpoints (GET, POST, PUT, DELETE) and handle request–response logic for the application. MongoDB would be used as the database to store and retrieve data, with components like routes, controllers, database connections, and basic validation forming the core of the API.
Evaluation: Score: 2 Reason: The answer correctly identifies Flask for handling API requests and responses and MongoDB for data storage. It also accurately lists the key components of such an API (routes, controllers, database connections, and validation). The brevity requirement is also met.

FINAL SCORE: 87/25